GLUING DIVIDED POWER STRUCTURES ON A SUM OF TWO IDEALS

THE STACKS PROJECT AUTHORS

Editorial note. No omission markers in-span; proof complete as written.

Allowed standard prerequisites (editorial).

- Definition of a divided power structure. definition-divided-powers (same chapter, definitional anchor). Axioms (1)-(5) used in the gluing verification.
- Generator criterion. lemma-check-on-generators (same chapter, named). Reduces the final axioms (2) and (5) for the glued structure to generators, where they are inherited from gamma and delta.

Source reference. Original definition 07GL: sources/stacks-dpa/pinned-original-source.tex, source lines 253–253. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 07GP: sources/stacks-dpa/pinned-original-source.tex, source lines 368–368. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let A be a ring with two ideals $I, J \subset A$. Let γ be a divided power structure on I and let δ be a divided power structure on J. Then*

(1) *γ and δ agree on IJ,*
(2) *if γ and δ agree on $I \cap J$ then they are the restriction of a unique divided power structure ϵ on $I + J$.*

Proof. Let $x \in I$ and $y \in J$. Then

$$\gamma_n(xy) = y^n\gamma_n(x) = n!\delta_n(y)\gamma_n(x) = \delta_n(y)x^n = \delta_n(xy).$$

Hence γ and δ agree on a set of (additive) generators of IJ. By property (4) of Definition 07GL it follows that they agree on all of IJ.

Assume γ and δ agree on $I \cap J$. Let $z \in I + J$. Write $z = x + y$ with $x \in I$ and $y \in J$. Then we set

$$\epsilon_n(z) = \sum \gamma_i(x)\delta_{n-i}(y)$$

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for all $n \geq 1$. To see that this is well defined, suppose that $z = x' + y'$ is another representation with $x' \in I$ and $y' \in J$. Then $w = x - x' = y' - y \in I \cap J$. Hence

$$\begin{aligned}
\sum\nolimits_{i+j=n} \gamma_i(x)\delta_j(y) &= \sum\nolimits_{i+j=n} \gamma_i(x' + w)\delta_j(y) \\
&= \sum\nolimits_{i'+l+j=n} \gamma_{i'}(x')\gamma_l(w)\delta_j(y) \\
&= \sum\nolimits_{i'+l+j=n} \gamma_{i'}(x')\delta_l(w)\delta_j(y) \\
&= \sum\nolimits_{i'+j'=n} \gamma_{i'}(x')\delta_{j'}(y + w) \\
&= \sum\nolimits_{i'+j'=n} \gamma_{i'}(x')\delta_{j'}(y')
\end{aligned}$$

as desired. Hence, we have defined maps $\epsilon_n : I + J \to I + J$ for all $n \geq 1$; it is easy to see that $\epsilon_n \mid_I = \gamma_n$ and $\epsilon_n \mid_J = \delta_n$. Next, we prove conditions (1) – (5) of Definition 07GL for the collection of maps ϵ_n. Properties (1) and (3) are clear. To see (4), suppose that $z = x + y$ and $z' = x' + y'$ with $x, x' \in I$ and $y, y' \in J$ and compute

$$\begin{aligned}
\epsilon_n(z + z') &= \sum\nolimits_{a+b=n} \gamma_a(x + x')\delta_b(y + y') \\
&= \sum\nolimits_{i+i'+j+j'=n} \gamma_i(x)\gamma_{i'}(x')\delta_j(y)\delta_{j'}(y') \\
&= \sum\nolimits_{k=0,\ldots,n} \sum\nolimits_{i+j=k} \gamma_i(x)\delta_j(y) \sum\nolimits_{i'+j'=n-k} \gamma_{i'}(x')\delta_{j'}(y') \\
&= \sum\nolimits_{k=0,\ldots,n} \epsilon_k(z)\epsilon_{n-k}(z')
\end{aligned}$$

as desired. Now we see that it suffices to prove (2) and (5) for elements of I or J, see Lemma 07GP. This is clear because γ and δ are divided power structures.

The existence of a divided power structure ϵ on $I + J$ whose restrictions to I and J are γ and δ is thus proven; its uniqueness is rather clear. □